

Tutorial 2, Friday, 09/10/2026, by vrm

Today: *LTL*, *CTL*, and *CTL**

Note. Unless stated otherwise, transition systems are finite Kripke structures and every state has at least one successor. LTL formulas are interpreted universally over all paths starting in the initial state.

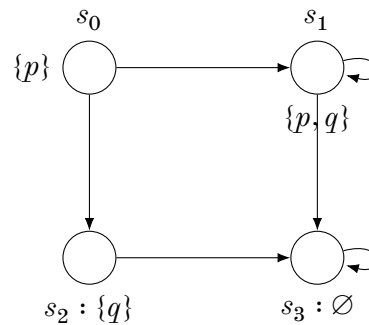
A. Reading and writing temporal formulas

Exercise 1. Let *req*, *ack*, *error*, *reset*, *crit* and *ready* be atomic propositions. Write formulas expressing the following properties. For each property, state whether your formula is in LTL, CTL, or requires CTL*.

- (a) Every request is eventually acknowledged.
- (b) An error never occurs.
- (c) From every reachable state, it is possible to reach a reset state.
- (d) Whenever the system enters a critical section, it eventually leaves it.
- (e) There exists a computation on which *ready* holds infinitely often.
- (f) Along every computation, from some point onward *ready* holds forever.
- (g) From every reachable state there exists a computation on which *ready* eventually holds forever.
- (h) Whenever *error* occurs, there exists a continuation on which *reset* is reached before the next occurrence of *error*.

B. Evaluate formulas on a transition system

Exercise 2. Consider the following Kripke structure.



For each formula below, determine the set of states satisfying it:

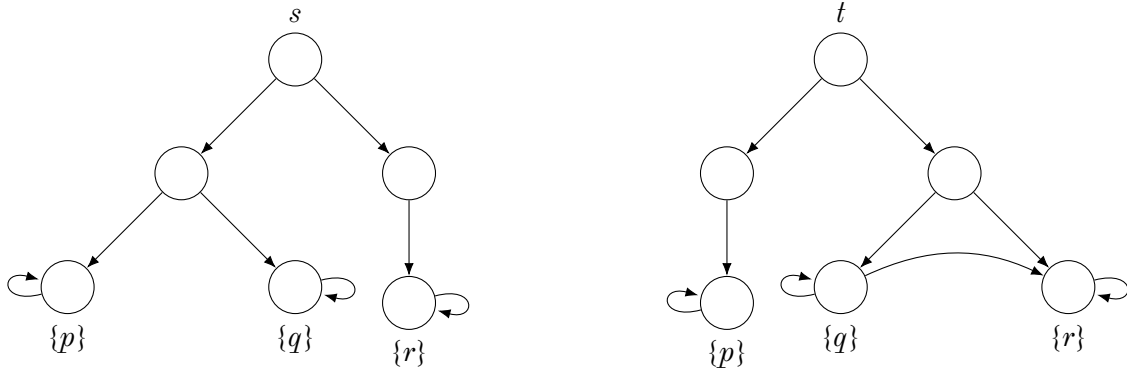
$$\text{Sat}(\varphi) = \{s \mid s \models \varphi\}.$$

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|------------|------------|----------------|
| (a) $EX q$ | (d) $AF q$ | (g) $E(p U q)$ |
| (b) $AX q$ | (e) $EG p$ | (h) $A(p U q)$ |
| (c) $EF q$ | (f) $AG p$ | (i) $AGEF q$ |

For at least two formulas, compute $\text{Sat}(\varphi)$ explicitly using the bottom-up CTL model-checking algorithm.

C. Find a distinguishing formula

Exercise 3. Consider the following two Kripke structures. All unlabeled states satisfy no atomic propositions.



- Find a CTL formula distinguishing s and t .
- Find such a formula of the smallest temporal depth you can.
- Show that the two systems do not have the same set of traces.
- Find an LTL formula distinguishing the two systems.
- Can an LTL formula distinguish the two systems when interpreted from s and t ? Justify your answer in terms of traces.

Exercise 4. Construct two finite Kripke structures with the same set of traces from their initial states, but which are distinguished by some CTL formula.

Your answer should contain:

- the two transition systems;
- an argument that their trace sets are equal;
- a CTL formula distinguishing their initial states.

D. Equivalences and counterexamples

Exercise 5. Prove the following LTL equivalences or give a counterexample if they are not valid.

- | | |
|---------------------------------------|---|
| (a) $FF\varphi \equiv F\varphi$ | (e) $X(\varphi \wedge \psi) \equiv X\varphi \wedge X\psi$, |
| (b) $GG\varphi \equiv G\varphi$, | (f) $F(\varphi \vee \psi) \equiv F\varphi \vee F\psi$, |
| (c) $FGF\varphi \equiv GF\varphi$, | (g) $G(\varphi \wedge \psi) \equiv G\varphi \wedge G\psi$. |
| (d) $GF G\varphi \equiv F G\varphi$, | |

For every proof, work directly from the semantics.

Exercise 6. For each proposed equivalence below, decide whether it is valid. If it is valid, prove it. Otherwise draw a counterexample.

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|-----|--|
| (a) | $F(\varphi \wedge \psi) \stackrel{?}{\equiv} F\varphi \wedge F\psi$ |
| (b) | $G(\varphi \vee \psi) \stackrel{?}{\equiv} G\varphi \vee G\psi$ |
| (c) | $EF(\varphi \vee \psi) \stackrel{?}{\equiv} EF\varphi \vee EF\psi$ |
| (d) | $EF(\varphi \wedge \psi) \stackrel{?}{\equiv} EF\varphi \wedge EF\psi$ |
| (e) | $AF(\varphi \wedge \psi) \stackrel{?}{\equiv} AF\varphi \wedge AF\psi$ |
| (f) | $EG(\varphi \wedge \psi) \stackrel{?}{\equiv} EG\varphi \wedge EG\psi$ |
| (g) | $AG(\varphi \wedge \psi) \stackrel{?}{\equiv} AG\varphi \wedge AG\psi$ |

Exercise 7. Prove the following CTL dualities:

$$\neg EX \varphi \equiv AX \neg \varphi,$$

$$\neg EF \varphi \equiv AG \neg \varphi,$$

$$\neg AF \varphi \equiv EG \neg \varphi.$$

Which of the following are valid?

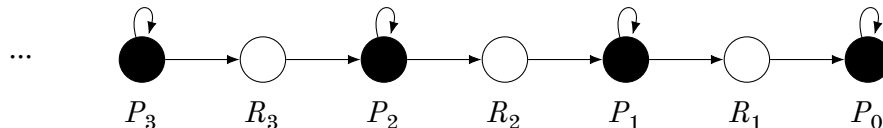
$$EX EF p \stackrel{?}{\equiv} EF EX p, \quad AX AF p \stackrel{?}{\equiv} AF AX p.$$

Give a proof or a counterexample.

E. Bounded CTL depth

Exercise 8. For $n \geq 0$, let $s \equiv_n t$ mean that s and t satisfy the same CTL formulas of temporal depth at most n .

Consider the infinite family below. Black states are P_i , white states are R_i .



Prove, by induction on n , the strengthened statement

$$P_k \equiv_n P_j, \quad R_k \equiv_n R_j \quad \text{whenever } n \leq k \leq j.$$

Your proof should explain why the statement has to be strengthened from just $P_n \equiv_n P_j$ to simultaneous statements about the P - and R -states.

Optional. Formulate the argument as an n -round game between two players.

F. LTL versus CTL

Exercise 9. The CTL* formula

$$AFG \text{ black}$$

is an LTL property under the usual universal interpretation of LTL.

Using the family from Exercise 11, construct two families of transition systems T_i and T'_i such that

$$T_i \models AFG \text{ black}, \quad T'_i \models AFG \text{ black},$$

but their initial states cannot be distinguished by CTL formulas of temporal depth at most i .

Conclude that the LTL property $FG \text{ black}$ is not expressible in CTL.

Exercise 10. Classify each of the following formulas as: LTL, CTL, CTL* — CTL.

- | | |
|-----------------------|--------------------------------|
| (a) $AGFp$ | (d) $A(p \cup (q \wedge EXr))$ |
| (b) $AGEFp$ | (e) $EFAGp$ |
| (c) $E(Fp \wedge Fq)$ | (f) $EG(Fp \vee Fq)$ |

For every CTL formula, rewrite it explicitly as a CTL* state formula.

G. Counterexamples and witnesses

Exercise 11. For finite Kripke structures, describe the form of a witness or counterexample for each of the following.

- | | |
|-----------|-------------------|
| (a) EFp | (d) AFp |
| (b) EGp | (e) $E(p \cup q)$ |
| (c) AGp | |

Show that whenever a finite Kripke structure violates AFp , there exists an ultimately periodic counterexample path (a finite prefix followed by a cycle).

H. Challenge: infinite-state systems

Exercise 12. [*Pushdown systems*] A pushdown system has configurations of the form (q, w) , where q is a control state and w is a stack word.

Investigate model checking for LTL and CTL under the following two choices of atomic propositions:

- (a) propositions depend only on the control state q ;
- (b) propositions are given by regular sets of configurations.

For every decidable case, describe the algorithm.

For every claimed undecidable variant, give a reduction or identify precisely which known undecidable problem is encoded.

Exercise 13. [*VASS / Petri nets*] Consider a VASS as an infinite transition system. Study the model-checking problems for LTL and CTL under state labellings defined by predicates on configurations.

Separate carefully the following questions:

- (a) propositions depending only on the finite control state;
- (b) reachability predicates;
- (c) coverability / upward-closed predicates;

For each setting, determine whether model checking is decidable. If you claim decidability, sketch an algorithm. If you claim undecidability, give a reduction.

I. Challenge: CTL* model checking

Exercise 14. Assume as a black box that for every LTL formula φ one can construct a Büchi automaton $\mathcal{A}_\varphi$ accepting exactly the words satisfying φ .

Design a model-checking algorithm for CTL* on finite Kripke structures.

- (a) Explain how to evaluate innermost CTL* state subformulas first.
- (b) Once their truth values are known, treat them as fresh atomic propositions.
- (c) Explain how an existential path formula $E\psi$ can be checked using a product with a Büchi automaton for ψ .
- (d) Explain how universal path formulas can be reduced to existential ones.

Give a rough upper bound on the complexity of your construction.